Index calculus for Curve25519: exact model binding and the remaining algebraic bottleneck

Codex research session

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Result

Curve25519 can now be fed **exactly** into this repository's short-Weierstrass Semaev machinery. The new Rust adapter carries RFC 7748's full 255-bit field, prime-order subgroup, base point, and a reversible Montgomery-to-short-Weierstrass translation; it also evaluates the exact third Semaev polynomial S_3. Six focused tests verify the map, the subgroup order, an S_3 zero-sum witness, the committed manifest, and a replayable identity certificate.

This does **not** produce a practical Curve25519 discrete-log attack. It makes the representation boundary exact and identifies the next real obstruction: constructing and solving a prime-field algebraic point-decomposition system at the required factor-base scale. In particular,

- $v_2(p-1) = 2$, so a binary Kummer base $g\mu_{2^t} \subset \mathbb{F}_p$ has at most $2^2 = 4$ abscissae;
- $v_2(p+1) = 1$, so the analogous binary norm-one/Dickson tower has at most 2 abscissae;
- auxiliary-isogeny and mixed-radix PKM constructions are not ruled out by those two facts, but the repository's end-to-end PKM harness is currently single-word ($p < 2^{63}$) and its measured solver degree is not bounded.

The correct status is therefore **exact adapter verified; full-size algebraic index calculus blocked at factor-base/solver integration**.

The editable source is `mechanism.dot`, and the vector render is `mechanism.svg`.

Question and scope

The question was: *what concrete index-calculus work can be completed for Curve25519 without presenting a toy curve or a coordinate change as an attack on the deployed curve?*

The exact target is the prime-order subgroup specified by RFC 7748:

$$E/\mathbb{F}_p : v^2 = u^3 + 486662u^2 + u, \qquad p = 2^{255} - 19,$$

with prime subgroup order

$$\ell = 2^{252} + \texttt{0x14def9dea2f79cd65812631a5cf5d3ed},$$

cofactor 8, and RFC base point $P = (u, v)$ with $u = 9$. This is an ECDLP model on $\langle P \rangle$, not an X25519 protocol transcript. X25519 uses only a u-coordinate and a clamped scalar; conflating that interface with an ECDLP instance would obscure the subgroup and sign information needed by relation collection.

No timing or scaling experiment was launched. The work consists of literature review, exact arithmetic, executable identities, and unit/property checks.

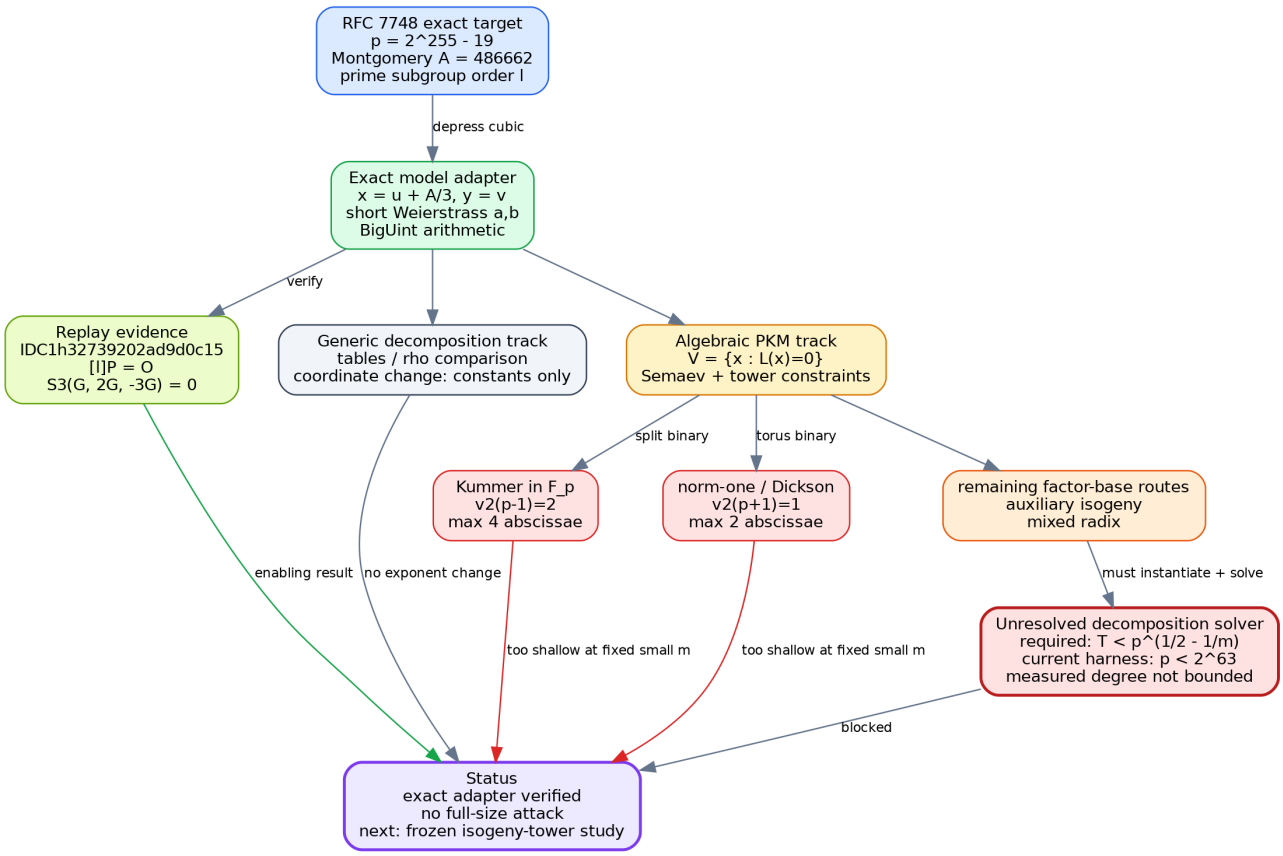


Figure 1: Curve25519 index-calculus mechanism and boundary

Literature boundary

The literature was reviewed before choosing the implementation.

1. **Semaev (2004)** introduced summation polynomials and the point-decomposition formulation. The local literature record was used for the foundational mechanism; its full PDF was not accessible in this session, so this report does not rest a new complexity claim on uninspected details.
2. **Petit, Kosters, and Messeng (PKC 2016)** give the primary prime-field algebraic factor-base construction used here as the serious lane. Their factor base is $F = \{P : x(P) \in V\}$, with $V = \{x : L(x) = 0\}$ and L a composition of low-degree maps. Keeping one auxiliary variable per map level avoids replacing the tower by one dense high-degree equation.
3. **Amadori, Pintore, and Sala (2018)** reduce their variant to one polynomial system, but explicitly do not claim a sub-$p^{1/2}$ algorithm; their experiments cover only 11–22-bit prime fields and leave the system-solving cost unresolved.
4. The repository's existing PKM tower study measures a growing solving degree: $m = 2$ reaches degree 6, $m = 3$ reaches 7, and the $m = 4, N = 16$ systems require roughly an hour and up to 8.1 GiB in the recorded round. Those are scaled systems, not Curve25519 measurements, but they directly reject the optimistic assumption that the current solver degree is already known to be bounded.

PKM's partial cost analysis makes the missing condition explicit. With $(\deg L)^m \approx m!p$, the decomposition solver $T(E, m, L)$ must beat $p^{1/2-1/m}$ (up to the paper's other terms) before the construction can beat a generic square-root method. An exact model conversion changes constants and representation; it does not establish that solver bound.

Exact model adapter

For a Montgomery cubic

$$v^2 = u^3 + Au^2 + u,$$

set

$$x = u + A/3, \quad y = v,$$

over a field of characteristic other than 3. Substitution gives

$$y^2 = x^3 + ax + b, \quad a = 1 - A^2/3, \quad b = 2A^3/27 - A/3.$$

The implementation computes every division as a modular inverse in $\mathbb{F}_p$. It does not round or truncate the 255-bit values. The native manifest `model.json` records the resulting a, b, translated base point, subgroup, and implementation boundary.

The integer identity behind the substitution is

$$\begin{aligned} 27(u^3 + Au^2 + u) = (3u + A)^3 \\ + 3(3 - A^2)(3u + A) + 2A^3 - 9A. \end{aligned}$$

Certificate `IDC1h32739202ad9d0c15` checks this degree-3 identity at 32 points derived from the sealed statement and at 32 additional domain-separated replay points. The checker reports an accept verdict and a Schwartz–Zippel false-accept bound below 2^{-3802} for that replay. This is an evaluated certificate, not a formal proof pointer; the displayed expansion is also directly inspectable.

Exact S_3 binding

For the translated curve $y^2 = x^3 + ax + b$, the implementation evaluates

$$\begin{aligned} S_3(x_1, x_2, x_3) = {} & (x_1 - x_2)^2 x_3^2 \\ & - 2\big((x_1 + x_2)(x_1 x_2 + a) + 2b\big)x_3 \\ & + (x_1 x_2 - a)^2 - 4b(x_1 + x_2) \pmod{p}. \end{aligned}$$

The focused test constructs $G, 2G, -3G$ on the exact translated curve and checks $S_3(x(G), x(2G), x(-3G)) = 0$, including a permuted evaluation. A separate test checks $[\ell]G = \mathcal{O}$. Together these checks bind the RFC point, the coordinate translation, the short-Weierstrass group law, and S_3 on the full-size field.

Curve25519-specific factor-base audit

For the binary Kummer construction used by the existing `prime-kummer` plug-in, $2^t \mid p - 1$ is required. Since

$$p - 1 = 2^2(2^{253} - 5),$$

the exact maximum is $t = 2$, hence four candidate abscissae before testing which ones lift to curve points. The binary norm-one/Dickson analogue uses the two-primary part of $p + 1$; here $v_2(p + 1) = 1$, hence two candidates.

For comparison, the PKM balance $|V|^m \approx m!p$ asks for approximately 2^{128}, $2^{85.9}$, or $2^{64.9}$ abscissae at $m = 2, 3, 4$, respectively. Four is therefore not a factor base for the usual small, fixed-m Curve25519 attack model.

This audit is deliberately narrow:

- it does not rule out a mixed-radix composition using odd factors;
- it does not rule out PKM's auxiliary-curve/isogeny construction, which does not require $2^t \mid p \pm 1$;
- it does not show that either remaining construction has an affordable precomputation or solver at Curve25519 scale.

Evidence and status

Requirement	Evidence	Status
Exact standard curve and subgroup	RFC 7748 parameters in `model.json`	Verified against source and native constants
Montgomery/short equivalence	inverse-map tests plus `IDC1h32739202ad9d0c15`	Verified
Correct subgroup generator	full-size test $[\ell]G = \mathcal{O}$	Verified
Semaev binding on the exact curve	$G, 2G, -3G$ zero-sum test	Verified
Binary PKM capacity	exact $v_2(p-1) = 2, v_2(p+1) = 1$ tests	Verified, narrow scope
Full relation collection and linear algebra	no full-size harness	Blocked
Solver beats rho	no such evidence	Not established
Performance claim	no experiment run	Not claimed

The exact replay commands are:

```
cargo check --example curve25519_ic_artifact
cargo test --lib curve25519_ic::tests -- --nocapture
cargo run --quiet --example curve25519_ic_artifact -- \
  replay research/curve25519_index_calculus_20261102/montgomery-translation.identity.json
```

Observed focused result: 6 passed, 0 failed; certificate replay accepted with 32 original and 32 extra points. The initial monolithic test link took 4m46s in this cloud environment; the tests themselves completed in 0.02s. That build time is environment context, not an algorithm measurement.

Next research step

The next useful step is not another coordinate model. It is a frozen, bounded **exact-versus-scaled isogeny-tower integration study**:

1. define an exact Curve25519 auxiliary-isogeny factor-base descriptor without enumerating its full set;
2. bind it to a scaled family whose tower equations have the same degrees and variable ordering;
3. pre-register (m,t), solver caps, null controls, and exact extrapolation rules;
4. measure solving degree and memory on the scaled family;
5. promote nothing to Curve25519 unless the exact descriptor and the scaling law both survive review.

This preserves the repository's separation between exact-curve facts, scaled measurements, and extrapolation. It also targets the one route not closed by the $p \pm 1$ audit.

Sources and local evidence

- A. Langley, M. Hamburg, and S. Turner, RFC 7748: Elliptic Curves for Security, Section 4.1. Exact Curve25519 parameters and base point.
- I. Semaev, Summation polynomials and the discrete logarithm problem on elliptic curves, 2004. Foundational summation-polynomial mechanism; full text was not retrieved in this session.
- C. Petit, M. Kosters, and A. Messeng, Algebraic approaches for the elliptic curve discrete logarithm problem over prime fields, PKC 2016; author PDF. Primary factor-base construction and partial cost analysis; full text read.
- A. Amadori, F. Pintore, and M. Sala, On the discrete logarithm problem for prime-field elliptic curves, *Finite Fields and Their Applications* 51 (2018), 168–182. Accepted manuscript read from the TU/e repository.
- K. Yokoyama, M. Yasuda, Y. Takahashi, and J. Kogure, Complexity bounds on Semaev's naive index calculus method for ECDLP, *Journal of Mathematical Cryptology* 14 (2020), 460–485. Bibliographic record/abstract only in this session; no detailed theorem from it is newly asserted here.
- Local measured context: `RESEARCH_PKM_TOWER_ORACLE.md`, `DERIVED.md`, and `DECISIONS.md`.

`ecdb` was not exposed in this environment, so standard parameters were taken from RFC 7748 and new values were labelled as native derivations. The canonical mechanism graph was checked conceptually and left unchanged: this study binds an existing PKM/Semaev mechanism to one standard curve; it does not introduce a new graph node or a measured edge.